Identity Management & Remote Access Configuration

Name: Hana
Reg.No: SAB01-4608
System Administration & Infrastructure
ITSimplera Solutions
Instructor: Tahir (Senior System Administrator)

Table Of Contents:

Introduction

Identity and Access Management (IAM) is one of the core responsibilities of every System Administrator. Managing user accounts, securing user access, configuring firewalls, and enabling secure remote administration are essential skills in enterprise IT environments.

Objective

This report provides a brief guide for configuring IAM, creating different types of user accounts, how to configure firewalls, and how to enable RDP.

Theory Topics

- IAM: identity and access management -> provide secure access to compnay's resources
- Standard User Accounts in Windows: regular account with restricted useage/no privileg
- Types of User Accounts (Administrator vs Standard User)
    - Standard User: only perform actions within the user profile.
    - Administrator User: has full control over the machine and unrestricted privileges.
- User Account Properties and Password Policies: configurations and settings that defines user's identity, settings, and access levels.
- Basic User Account Troubleshooting
- Windows Firewall Fundamentals : built-in security feature that filters network traffic moving in and out of the machine.

- Firewall Rules (Inbound & Outbound)
    - Inbound traffic: originate from outside the network, such as an external user with a web browser, email client, server or application making requests.
    - Outbound traffic: originate from inside the network, destined for services on the internet or outside networks, such as a user visiting an external website or an internal mail server connecting to an external one.
- Introduction to Remote Desktop Protocol (RDP): enables users to control and operate computers from a distance remotely.

Software Requirements

Operating System: Windows 10 VM
Hypervisor: VMware Workstation
Access Level: Local Administrator Account
Connectivity: Internet Connection

Implementation

Creating and managing two standard local accounts

Using following commands to create and configure accounts

Create a new user with name and password

```
net user {username} {password} /add.
```

Set other details

```
net user User_Alpha /fullname:"Alpha Smith"
/comment:"Standard local account for testing"
```

Set password expiration

```
wmic useraccount where name='User_Alpha' set
PasswordExpires=FALSE
```

verify details

```
net user {username}
```

list all local users

```
net user
```

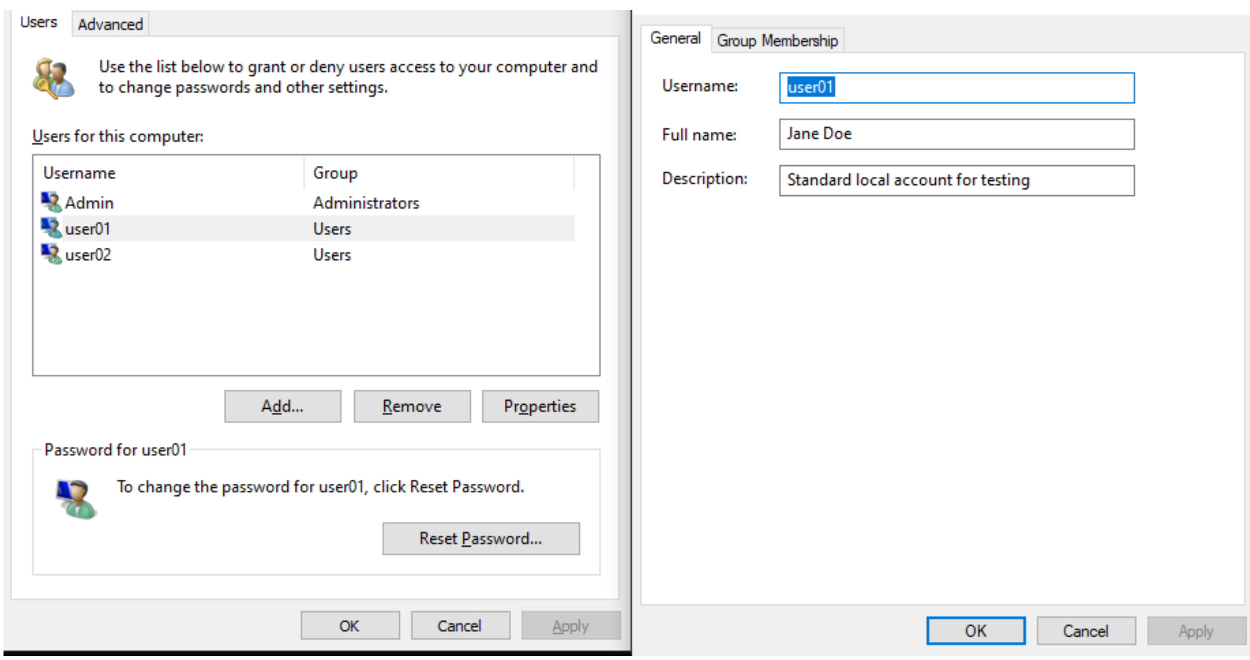
Accounts details:

```
C:\Windows\system32>net user user02 00001111 /add
The command completed successfully.


C:\Windows\system32>net user user02 /fullname:"Joe Doe" /comment:"Standard local account for testing"
The command completed successfully.


C:\Windows\system32>wmic useraccount where name="user02" set PasswordExpires=FALSE
Updating property(s) of '\\DESKTOP-IJ4IOSM\ROOT\CIMV2:Win32_UserAccount.Domain="DESKTOP-IJ4IOSM",Name="user02"'
Property(s) update successful.
```

```
C:\Windows\system32>net user user01
User name                    user01
Full Name                    Jane Doe
Comment                      Standard local account for testing
User's comment
Country/region code          000 (System Default)
Account active               Yes
Account expires              Never

Password last set            08/07/2026 15:22:47
Password expires             Never
Password changeable          08/07/2026 15:22:47
Password required            Yes
User may change password     Yes

Workstations allowed         All
Logon script
User profile
Home directory
Last logon                   Never

Logon hours allowed          All

Local Group Memberships      *Users
Global Group memberships     *None
The command completed successfully.
```

```
User name                    user02
Full Name                    Joe Doe
Comment                      Standard local account for testing
User's comment
Country/region code          000 (System Default)
Account active               Yes
Account expires              Never

Password last set            08/07/2026 15:30:34
Password expires             Never
Password changeable          08/07/2026 15:30:34
Password required            Yes
User may change password     Yes

Workstations allowed         All
Logon script
User profile
Home directory
Last logon                   Never

Logon hours allowed          All

Local Group Memberships      *Users
Global Group memberships     *None
The command completed successfully.
```

Can also be created using GUI



Applying Local Security Policies

Policy	Security Setting
Enforce password history	0 passwords remembered
Maximum password age	42 days
Minimum password age	0 days
Minimum password length	0 characters
Minimum password length audit	Not Defined
Password must meet complexity requirements	Disabled
Relax minimum password length limits	Not Defined
Store passwords using reversible encryption	Disabled

- Minimum password length
    - Purpose: enforce minimum password length
    - Importance: increased security, Weak, short passwords are easily cracked
- Enforce password history
    - Purpose: remember the last x passwords
    - Importance: increased security: users can't just reuse the same one

Policy	Security Setting
Enforce password history	3 passwords remembered
Maximum password age	42 days
Minimum password age	0 days
Minimum password length	8 characters
Minimum password length audit	Not Defined
Password must meet complexity requirements	Disabled
Relax minimum password length limits	Not Defined
Store passwords using reversible encryption	Disabled

Policy	Security Setting
Account lockout duration	10 minutes
Account lockout threshold	10 invalid logon attempts

- Account Lockout threshold

- Purpose: defines the exact number of failed password attempts a user is allowed to make before Windows completely locks the account.
- Importance: defense against automated guessing attacks

Policy	Security Setting
Account lockout duration	10 minutes
Account lockout threshold	5 invalid logon attempts

Interactive logon: Display user information when the session...	Not Defined
Interactive logon: Do not require CTRL+ALT+DEL	Not Defined
Interactive logon: Don't display last signed-in	Disabled
Interactive logon: Don't display username at sign-in	Not Defined

- Interactive logon: don’t display last signed-in

Purpose: When a user locks or logs out of the PC, Windows won't show the username of the last person who used it.

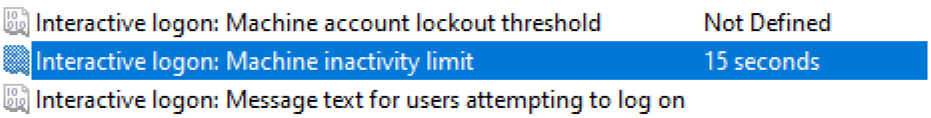
Importance: increased security: A hacker or an unauthorized person has to guess both the username and the password, not just the password.

Interactive logon: Display user information when the session...	Not Defined
Interactive logon: Do not require CTRL+ALT+DEL	Not Defined
Interactive logon: Don't display last signed-in	Enabled
Interactive logon: Don't display username at sign-in	Not Defined

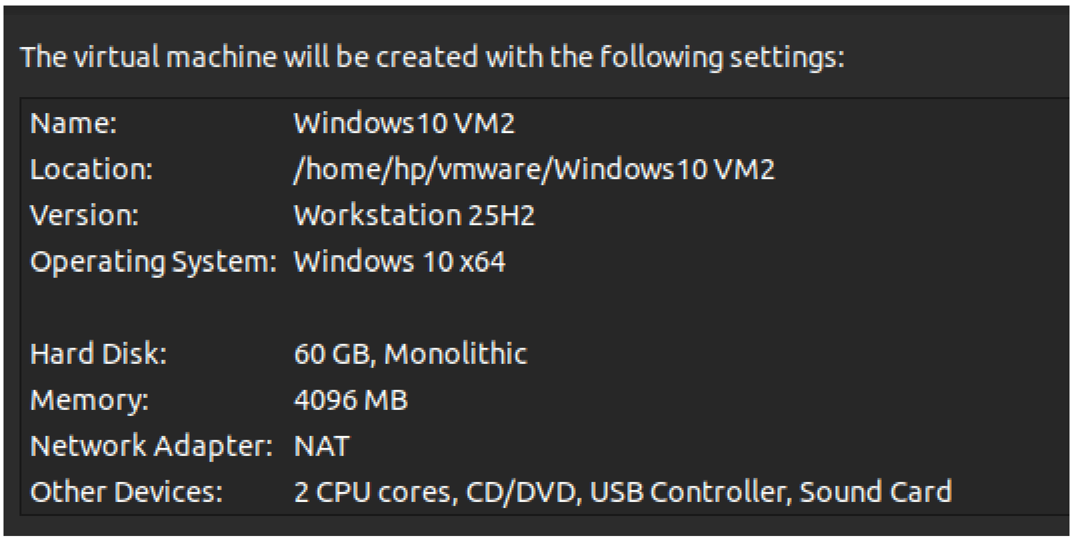
Interactive logon: Machine account lockout threshold	Not Defined
Interactive logon: Machine inactivity limit	Not Defined
Interactive logon: Message text for users attempting to log on	

- Machine inactivity limit
    - Purpose: automatically lock the screen and require a password if the computer is left idle for a specific number of seconds.

- Importance: increased security -> unauthorized physical access.

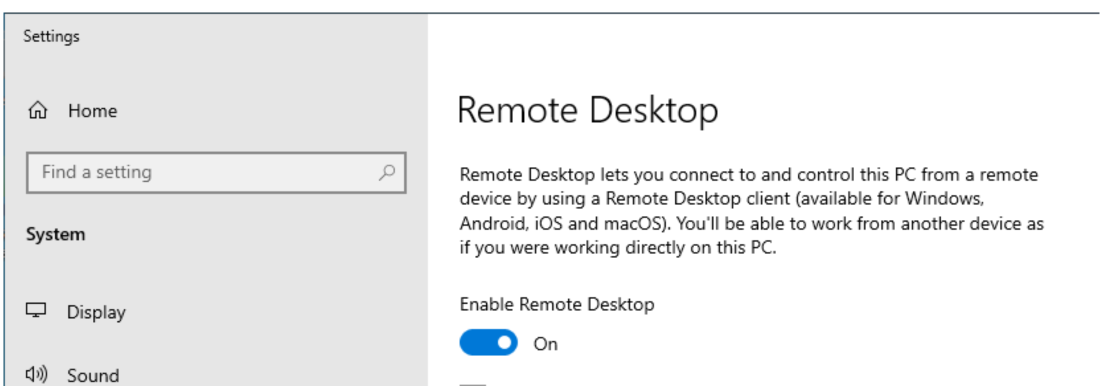


Deploying a second VM



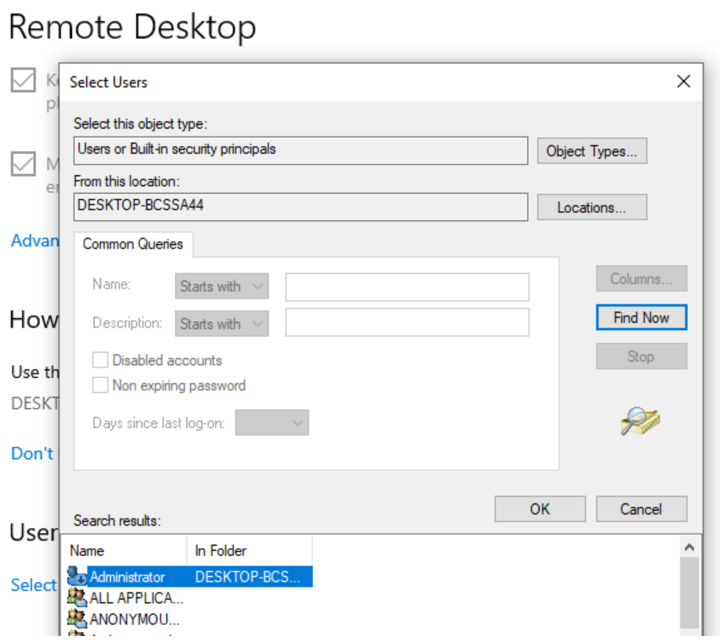
Remote Desktop (RDP) Configuration

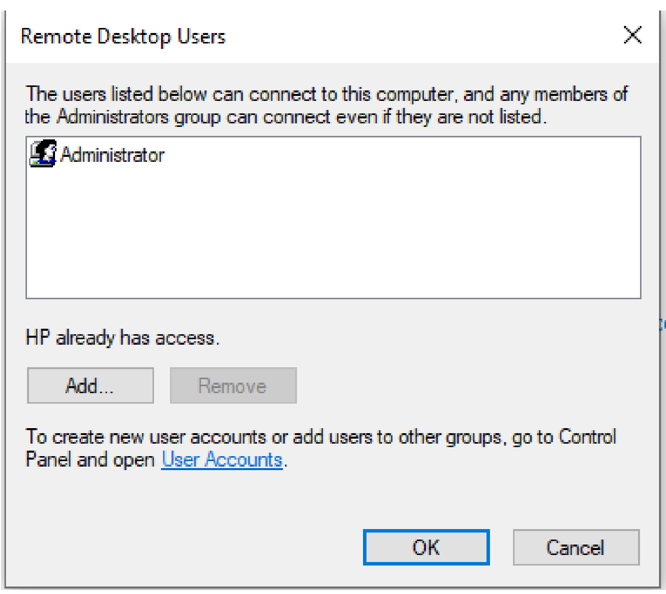
1. Enabling remote desktop on target machine



2. Configure Remote Desktop permissions : to define who is authorized to log in remotely to that specific device (add specific

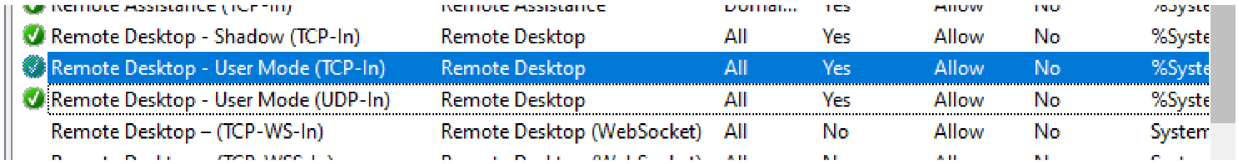
user from the target VM, to authorize it to receive remote connections)





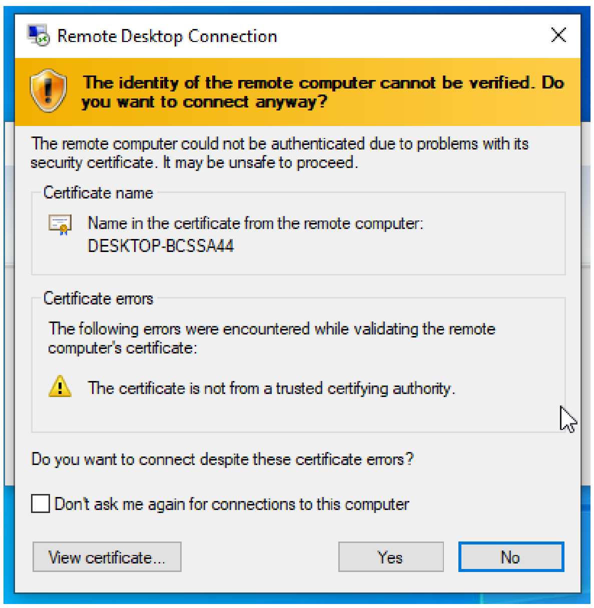
3. Allow the required Windows Firewall settings

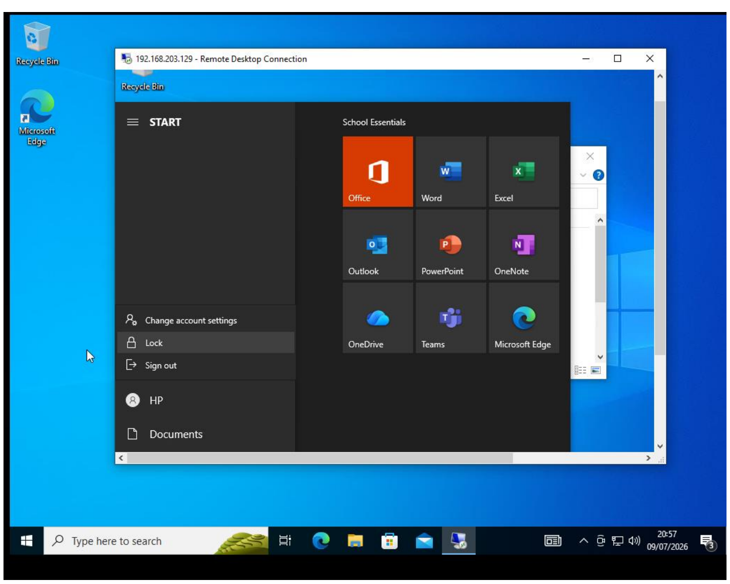
Windows Defender Firewall with Advanced Security -> Inbound Rules -> Remote Desktop - User Mode (TCP-In) -> enable



4. Establishing connection from the Main VM

Remote Desktop Connection -> IPv4 address of the target VM -> connect -> credentials of the added user to the target machine (display name)





Issue encountered: reconnecting/opening target VM will disconnect the remote access; Windows Client editions (like Windows 10 or 11) are designed as "single-user" operating systems. To watch/observe (like screen share) we need to use different tools.

Custom Firewall Rules

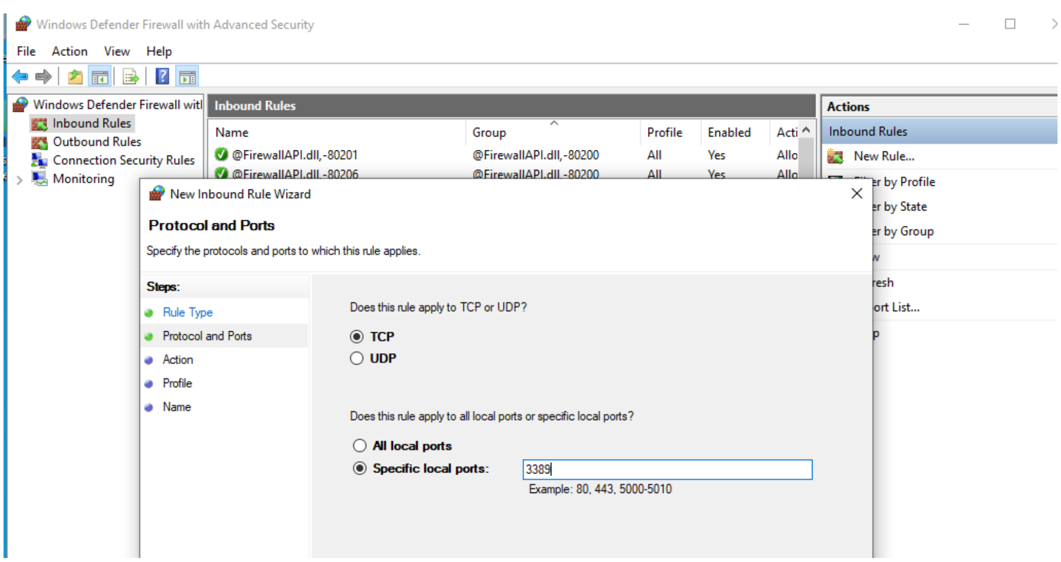
1. custom Inbound Rule that allows Remote Desktop traffic between the two virtual machines.

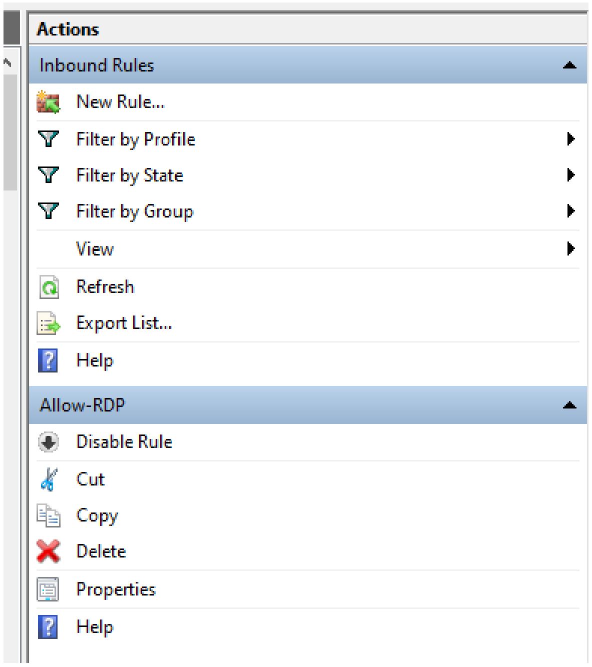
   Windows Defender Firewall with Advanced Security -> Inbound Rules -> New Rule (in Actions pane) ->

   - Rule type : Port
   - Protocols & Ports: TCP (local port 3389)
   - Action: Allow the connection

   -> check all Domain, Private, and Public

   - name : Allow-RDP





2. custom firewall rule that blocks Remote Desktop traffic.

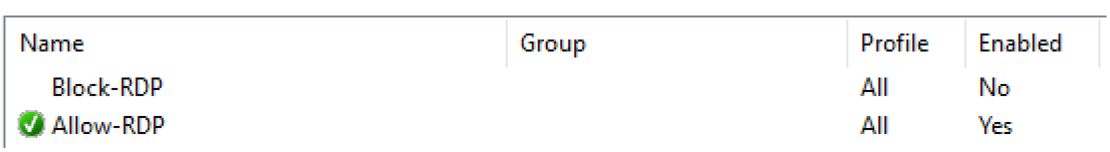
   Windows Defender Firewall with Advanced Security -> Inbound Rules -> New Rule (in Actions pane) ->

   - Rule type : Port
   - Protocols & Ports: TCP (local port 3389)
   - Action: block the connection

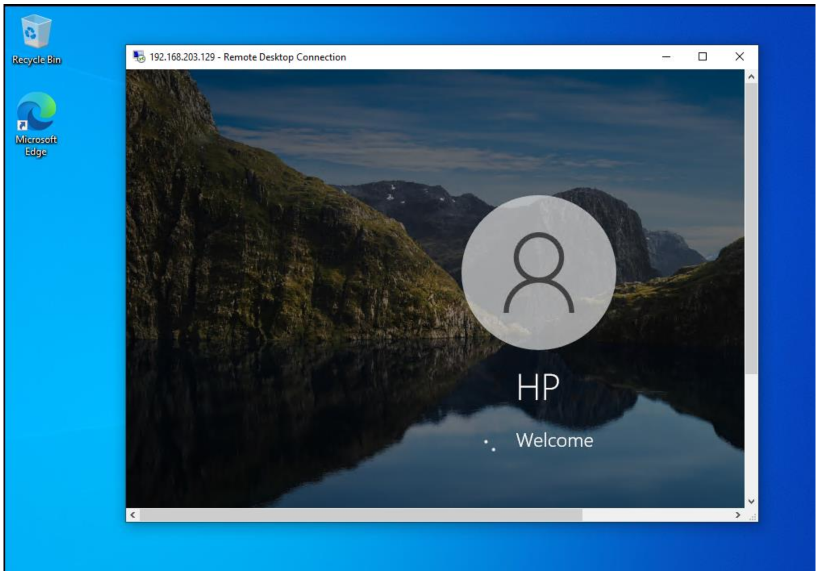
   -> check all Domain, Private, and Public

   - name : Block-RDP

- Successful RDP connection

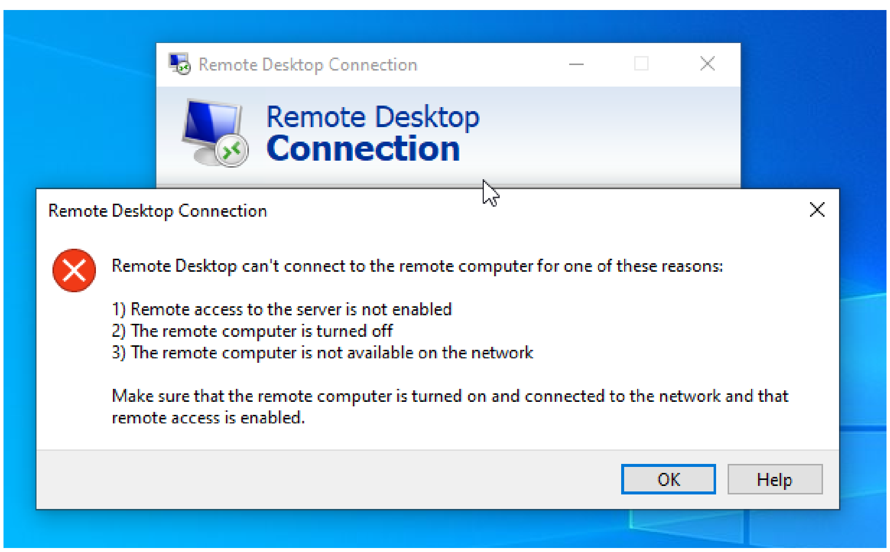


Name	Group	Profile	Enabled
Block-RDP		All	No
Allow-RDP		All	Yes



- Blocked RDP connection

Name	Group	Profile	Enabled
Block-RDP		All	Yes
Allow-RDP		All	No



Challenges faced

1. When successfully connecting from main to target VM, then opening the target VM, remote access will be disconnected; that’s because Windows Client editions (like Windows 10 or 11) are designed as "single-user" operating systems. To watch/observe (like screen share) we need to use different tools.
2. What will happen if both Allow-RDP & Block-RDP rules are enabled?

   -> the "Block" rule will always win.

   This is a fundamental principle of Windows Firewall (and almost all professional firewalls): Block rules take precedence over Allow rules.

Learning outcomes

Practical experience in creating and managing local user accounts, securely configuring remote access on a Windows 10 machine, applying security policies, and creating custom firewall rules.

Conclusion

This task demonstrated essential skills in managing Windows user accounts, security policies, and remote access.

References

- https://www.geeksforgeeks.org/blogs/how-to-create-a-new-user-in-windows-10/
- https://gaoamanda40.medium.com/local-security-policy-windows-10-what-is-it-how-to-open-it-17286b69f45
- https://www.microsoft.com/en-us/security/business/security-101/what-is-identity-access-management-iam
- https://netwrix.com/en/resources/blog/active-directory-password-policy/
- https://www.techtarget.com/searchsecurity/answer/Comparing-firewalls-Differences-between-an-inbound-outbound-firewall